SAFE-FOOD

MSc Knowledge Representation / Verified Windows Docker run / 20 October 2026

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Scope and integrity. This is an educational, synthetic two-source knowledge graph using OFF-shaped and USDA-shaped records. None of the source rows, shared barcodes or nutrition labels is independently verified against real products. It is not a food-safety or clinical decision system.

Executive summary

SAFE-FOOD prepares two heterogeneous teaching datasets, integrates them into an RDF/OWL application graph, validates them with SHACL, exposes the result in Apache Jena Fuseki, and answers selected English questions via generated SPARQL. It separately tests an unchanged GraphSAGE-style learning model against logistic regression. Windows/Docker operation, seven project tests, and an opt-in live Gemini micro-evaluation were completed.

Verified measure	Observation
Prepared synthetic records	2,925 OFF-shaped + 797 USDA-shaped
RDF knowledge graph	3,722 typed products / 56,474 triples
Exact-barcode candidate identity links	146 pairs / 292 directed sameProductAs triples
Prepared graph SHACL	0 violations / 0 warnings
Project tests	7 passed, 0 skipped; 2 non-fatal serializer warnings
Live Gemini schema QA, final targeted run	F1 1.000 on 5 answerable questions; 1/1 correct refusal

1 / Data and preparation

Raw synthetic fixtures contain 3,001 OFF-shaped rows and 800 USDA-shaped rows, with deliberate duplicates, invalid entries and planted barcode overlaps. Cleaning leaves 2,925 OFF and 797 USDA product rows and a traceable audit under data/quality/. Missing optional values remain missing rather than imputed to zero. SQLite creates 4,264 declared-allergen link rows.

2 / Ontology, mapping and constraints

foodsafe.ttl declares product, allergen, data-source and nutrition concepts. foodon-slim.ttl reuses a limited identifier subset, not the complete FoodOn ontology. The Python RDF materializer constructs 56,474 triples; mapping.ttl describes an R2RML counterpart. Source-specific nodes remain separate, even when an exact barcode generates symmetric sameProductAs links. Zero SHACL violations apply only to the stated prepared-data constraints, not to factual correctness.

3 / Fuseki and SPARQL

A running Fuseki kr dataset returned 3,722 products and 146 OFF-to-USDA matched candidate pairs. A second query demonstrated 333 peanut-allergen declarations and 16 distinct associated brand-name values. The first live peanut query failed because the model used AllergenPeanut instead of the graph individual AllergenPeanuts; the corrected query returned all 16 brand names.

4 / Unchanged graph-learning experiment

The classification task uses 758 positive (Nutri-Score A/B), 1,187 negative (C/D/E) and 1,777 unlabelled nodes. Five nutritional features are used: energy, fat, sugars, protein and salt. The GraphSAGE-style model aggregates same-brand neighbors, not the sameProductAs or product-allergen relations. Train-only imputation/scaling and brand-disjoint splits were used at seeds 11, 23 and 37.

Seed	Logistic baseline F1	GraphSAGE-style F1
11	0.5062	0.5211
23	0.4830	0.4957
37	0.5293	0.5038
Mean	0.5062	0.5069

The mean difference (~0.0007 Macro-F1) is negligible. Synthetic Nutri-Score labels lack the realism needed to claim nutritional predictive validity. Allergen link ranking separately gave approximate mean MRR 0.2573 (popularity) versus 0.2519 (brand-neighbor heuristic); this was not a neural link-prediction model.

5 / Access layer and error analysis

Offline mode supplies scripted demo answers and cannot be presented as LLM performance. After enabling live Gemini privately, the source-pair count query returned 146. On a six-question live micro-suite (five answerable, one medical refusal), the initial schema setup reached mean F1 0.800; examples and repair settings were 0.688 each. The peanut question alone scored zero because an unknown individual IRI was not rejected. After targeted exact-IRI prompt guidance and an allergen-individual validator, the schema-only rerun reached mean F1 1.000 and one correct refusal. This was not a broad held-out study.

A controlled mock model test supplied the wrong allergen IRI on attempt 1. The validator returned a correction hint, and attempt 2 executed a valid query returning 16 brands. This verifies the simulated repair loop only; no live Gemini repair was required or observed in the final evaluation.

6 / Quality of evidence and outstanding checks

- Verified: Windows Docker build, RDF/SHACL pipeline, Fuseki SPARQL, learning outputs, 7/7 unit tests, live Gemini query/evaluation and mocked repair control flow.
- Not verified: independent Morph-KGC/Python mapping parity; complete Protege reasoner consistency; a new-machine reproduction; real OFF/USDA identity precision; large held-out Gemini QA generalization.
- Security: the Google API key stays in a local .env ignored by Git; the report and source archive contain no credential values.

Conclusion

SAFE-FOOD demonstrates a runnable ontology-backed synthetic two-source data integration architecture with tested query access and graph-learning experiments. It does not establish real-world product-safety accuracy or GNN superiority. An evidence-based next phase would collect verified OFF/USDA branded data and conduct independent entity-resolution and external evaluation.

Project authors: Mohammad Diab and Mohammad Mantach. Individual role allocations were not supplied; add only if required by the course. This verified report supersedes earlier PDF drafts.